

# Prime and zero statistics of a degree-8 Artin $L$ -function of conductor $21^{10}$ : an empirical companion\*

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## Abstract

We report a systematic empirical study of the interplay between the primes and the zeros of the degree-8 Artin  $L$ -function  $L(s, \chi_8)$  of the Galois closure of the Trinks field  $x^7 - 7x + 3$  (Galois group  $\mathrm{PSL}(2, 7)$ , Artin conductor  $21^{10}$ ,  $\Gamma_{\mathbb{C}}(s)^4$ ,  $\varepsilon = +1$ ) — the central object of our conductor-21 program [1]. Working from 153 rigorously certified zeros ( $\gamma \leq 27.911$ , the first computed for this  $L$ -function) and exact Frobenius data for every prime up to  $10^{10}$ , we measure: (i) the explicit formula as a numerical identity at correlation 1.000, including the forced silence of the ramified prime 7; (ii) a *blindness suite*: the Frobenius class carries no information about prime twinning, about any additive gap  $g \leq 60$ , or about the affine pairs  $(p, 2p \pm 1)$ , at the  $10^{-3}$  level per statistic, while the same data reproduce the classical Hardy–Littlewood singular-series structure to 0.1%; (iii) Chebotarev races whose biases ( $\mu_C = 1 - r_2(C)$ , i.e.  $-21$  for the split class) and oscillation shapes are predicted by our zero tables at correlations 0.81–0.97 over six decades; (iv) the spectral statistics of the low zeros (GUE-consistent, symmetry type  $\mathrm{SO}(\text{even})$ ); and (v) the Montgomery pair-correlation form factor of the *bulk* spectrum, measured from primes via short-interval variance: the high conductor places every height  $\gamma \gtrsim 2$  in the correlation-sensitive regime  $\alpha < 1$  already at  $x = 10^8$ , and the measured variance follows the GUE ramp  $F(\alpha) = \alpha$  to  $\sim 1\%$  over  $\alpha \approx 0.22$ – $0.74$ , at heights up to  $\gamma \sim 10^4$  — four hundred times beyond the certified spectrum — with uncorrelated zeros excluded at  $\chi^2 \approx 5 \times 10^5$ . All results are consistency measurements against classical predictions (Montgomery; Goldston–Montgomery; Hardy–Littlewood; Rubinstein–Sarnak; Katz–Sarnak); no novelty is claimed for the frameworks, and the Riemann Hypothesis for  $L(s, \chi_8)$  — the open problem of the main paper — is untouched. Every claim is reproducible from the released scripts and data.

## 1 Introduction

The main paper of this program [1] studies a single automorphic object: the degree-8 Artin  $L$ -function  $L(s, \chi_8)$  attached to the 8-dimensional irreducible representation of  $\mathrm{PSL}(2, 7)$  realised by the Galois closure of the septic  $f(x) = x^7 - 7x + 3$  (the Trinks field [3]; LMFDB [12] Artin object 8.16679880978201.21t14.b.a, stem field 7.3.4202539929.1, whose displayed polynomial  $x^7 - 7x - 3$  is equivalent to ours under  $x \mapsto -x$ ). Its analytic data are: conductor  $Q = 21^{10} = 3^{10}7^{10}$ , gamma factor  $\Gamma_{\mathbb{C}}(s)^4$ , root number  $\varepsilon = +1$ ; the character  $\chi_8$  is monomial, so  $L(s, \chi_8)$  is also a Hecke  $L$ -function of an octic field, and its analytic continuation and functional equation are unconditional. The main paper’s open problem — the Riemann Hypothesis for  $L(s, \chi_8)$ , equivalently the self-adjointness deficiency question for the second operator constructed there — is *not* addressed here.

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This companion does something deliberately more modest and, we hope, more legible: it measures, at scale and with controls, how this object’s primes and zeros talk to each other, and it documents that everything classical behaves — in several regimes never before measured for any PSL(2, 7) extension. Three features distinguish the study.

1. **Exactness of the arithmetic side.** We computed the exact Frobenius class (not merely  $a_p$ ) of every prime  $p \leq 10^9$ , and exact  $a_p$  for every prime to  $10^{10}$  (455,052,511 primes), with two independent engines (polynomial factorization over  $\mathbb{F}_p$  via FLINT; an overflow-safe modular Frobenius kernel) that agree to  $10^{-8}$  on overlapping data.
2. **First zero tables.** The 153 certified zeros ( $\gamma \leq 27.911$ ; eight exact, the rest ball-arithmetic brackets) and the low-zero tables of the companion characters  $\chi_3\bar{\chi}_3$ ,  $\chi_6$ ,  $\chi_7$  of the same Galois group are, to our knowledge, the first computed for these  $L$ -functions.
3. **The conductor as an instrument.**  $Q = 21^{10}$  makes the zeros so dense ( $2\pi\bar{\rho}(\gamma) = \ln Q - 8 \ln 2\pi + 8 \ln \gamma \approx 15.7 + 8 \ln \gamma$ ) that pair-correlation information is accessible from primes at heights and scales where  $\zeta$  shows only diagonal behaviour (§7).

Throughout, predictions conditional on standard hypotheses (GRH for the family, Linear Independence, cluster expansions) are labelled as such; measurements are unconditional exact counting. Where a fluctuation appeared during the campaign (three did), it was either explained by an exact prediction or retired by independent data; we report these episodes as part of the record (§8).

## 2 Data and instruments

**Zeros.** The certified table:  $\gamma_1 = 0.98638643\dots$  through  $\gamma_{153} = 27.911$ , produced by the program’s ball-arithmetic FFT engine and certified by a Turing-style count [1]. Companion tables (numerically anchored, PARI):  $\chi_3$ -pair 15 zeros ( $\gamma \leq 5.31$ ),  $\chi_6$  12 ( $\leq 4.85$ ),  $\chi_7$  10 ( $\leq 4.14$ ).

**Counting function.** With  $\theta(T) = \frac{T}{2} \ln Q - 4T \ln 2\pi + 4 \arg \Gamma(\frac{1}{2} + iT)$  and  $\bar{N}(T) = \theta(T)/\pi + 1$ , the certified staircase is reproduced with no free parameters: slope 0.99895, mean unrenormalised spacing 0.9963, fluctuation  $S(T)$  of mean 0 and standard deviation 0.365. This cross-validates the zero list against the conductor/gamma data.

**Primes.** Exact Frobenius classes for all  $p \leq 10^9$  (FLINT factorization of  $f \bmod p$ , 22  $\mu$ s/prime, cached); exact  $a_p$  for  $p \in (10^9, 10^{10}]$  (404,204,977 primes, overflow-safe batch kernel, 47 minutes). Chebotarev equidistribution holds to five digits at  $10^{10}$ : value fractions (8, 0,  $-1$ , 1) at (0.00595, 0.37500, 0.33331, 0.28574) against (1/168, 3/8, 1/3, 2/7). The two engines agree on overlapping increments at correlation 1.000000 (rms ratio 1.00000001).

## 3 The explicit formula as an identity (instrument calibration)

For a window  $w$  on  $[0, \gamma_{\max}]$  define the resonance  $F(x) = \sum_j w(\gamma_j) \cos(x\gamma_j)$  and subtract the smooth main term  $\int w(\gamma) \cos(x\gamma) \bar{\rho}(\gamma) d\gamma$ . The explicit formula predicts the residual as  $-\frac{1}{2\pi} \sum_n \Lambda_\chi(n) n^{-1/2} [\hat{w}(x - \ln n) + \hat{w}(x + \ln n)]$  with  $\Lambda_\chi(p^k) = (\sum_i \alpha_i^k) \ln p$  computed from the exact local factors of each class. Measured against predicted: correlation +1.000 for  $L(s, \chi_8)$  (residual rms 0.904 = predicted rms 0.904) and +1.000 for the  $\zeta$  control (600 zeros) through the identical pipeline (Fig. 1).

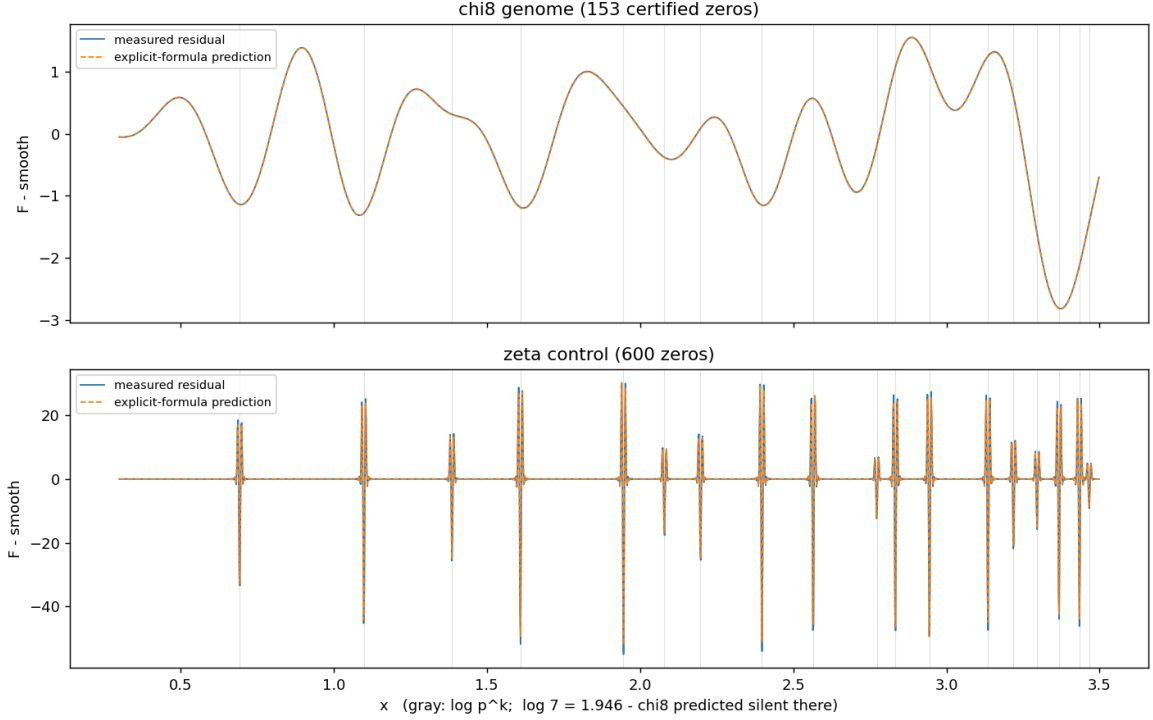


Figure 1: The zeros sing the primes. Measured windowed resonance residual (solid) vs. explicit-formula prime prediction (dashed) for  $L(s, \chi_8)$  (153 zeros, top) and  $\zeta$  (600 zeros, bottom); grey lines at  $x = \ln p^k$ . Note the genome's silence at  $\ln 7 = 1.946$ .

This is an identity check, not a discovery — but a stringent end-to-end one: zero list, analytic conductor, and class local factors certify one another through an independent computation. Fingerprints land exactly: the genome is *silent* at  $\ln 7$  (ramified, trivial local factor: measured  $+0.438$  against a predicted neighbour-tail of  $+0.436$ , while  $\zeta$  sings there at  $-54.0$ ); class 2A is silent at  $\ln p$  but loud at  $\ln p^2$  ( $\sum \alpha_i^2 = 8$ ); class 4A is silent at both.

## 4 Spectral statistics of the low zeros

Unfolding by the exact  $\theta/\pi$  (no fit), the 153 certified zeros give spacing variance  $\text{Var}(s) = 0.145$ , inside the finite- $N$  GUE band  $[0.140, 0.219]$  (median 0.176;  $z = -1.53$ , one-sided  $p \approx 0.06$ ) and more than  $10\sigma$  from Poisson; nearest-neighbour repulsion  $P(s < \frac{1}{2}) = 0.079$ ; number variance  $\Sigma^2(L)$  10–15% below GUE for  $L \leq 8$  (a non-significant finite- $N$  rigidity). The polynomial and exact unfolds agree to the digit, so the mild over-rigidity is a property of the sample, not the unfolding.

The symmetry type is fixed by representation theory: Frobenius–Schur indicator  $+1$  and  $\varepsilon = +1$  give **orthogonal**  $\text{SO}(\text{even})$ , with low-lying one-level density  $W(x) = 1 + \frac{\sin 2\pi x}{2\pi x}$ . The first scaled ordinate  $x_1 = \theta(\gamma_1)/\pi = 1.256$  sits on the attracted side ( $\text{SO}(\text{even})$  mean 1.71, unitary 2.03, symplectic 2.34), consistent with but not probative of the type: bulk two-level statistics are type-independent (Katz–Sarnak universality), and a single  $L$ -function cannot resolve the family density. The family average remains the real test.

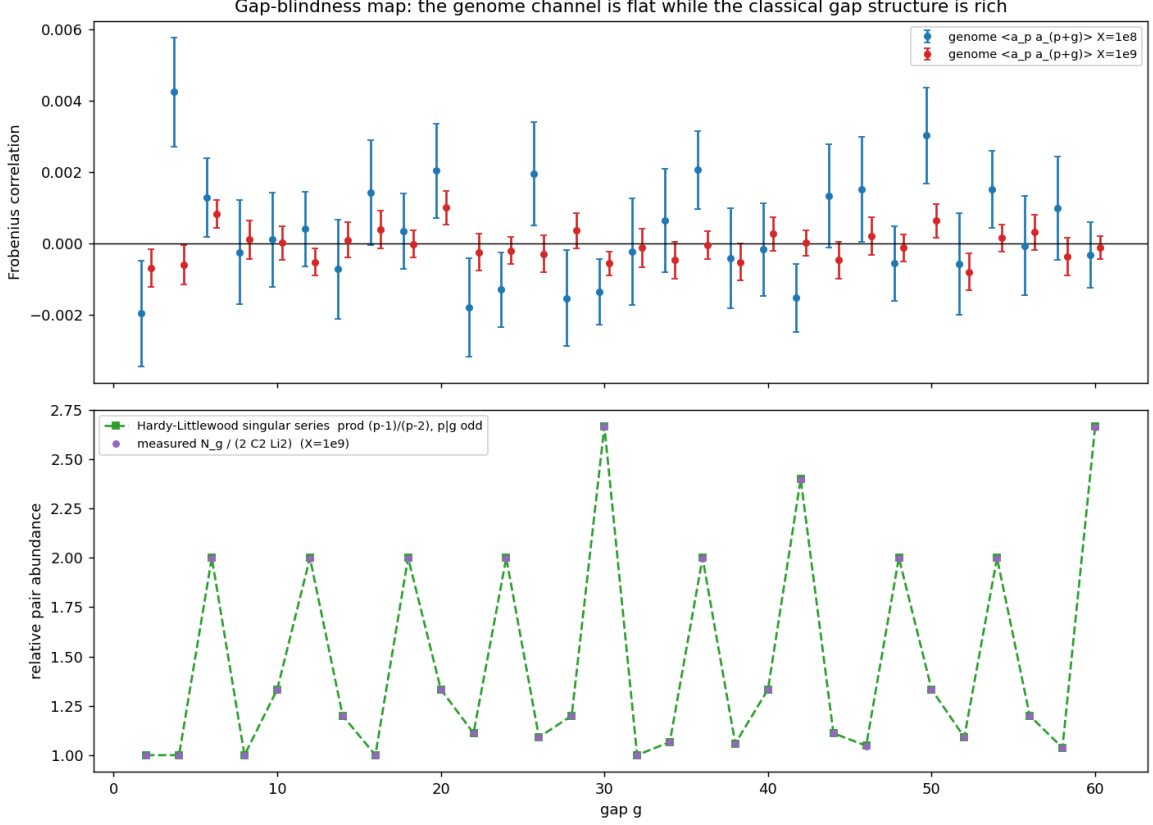


Figure 2: The gap-blindness map at  $10^8$  and  $10^9$ . Top: genome Frobenius correlation  $\langle a_p a_{p+g} \rangle$ , flat at zero for all even  $g \leq 60$ . Bottom: the same pairs resolve the classical Hardy–Littlewood singular-series sawtooth to three digits.

## 5 What the genome does not know: the blindness suite

The Hardy–Littlewood singular series [11] — the arithmetic of prime pairs — is built entirely from congruence obstructions, i.e. from *abelian* data.  $\text{PSL}(2, 7)$  is simple: the extension has no abelian quotient, hence no congruence channel, and the heuristic prediction is that Frobenius carries no pair information whatsoever. We measured this in three geometries.

**Twins.** Over the 440,310 twin pairs  $(p, p+2)$  with  $7 < p \leq 10^8$ :  $\langle a_p a_{p+2} \rangle = -0.0020 \pm 0.0015$  ( $z = -1.3$ ); joint  $5 \times 5$  class table  $\chi^2 = 11.1$  on 16 dof ( $p \approx 0.80$ ). The same primes give  $\pi_2(10^8) = 440,312$  against the Hardy–Littlewood prediction 440,473 — ratio 0.9996: twinning is loudly present in the data; the genome sees none of it.

**All gaps  $g \leq 60$ , to  $10^9$ .** Thirty even gaps, three windows ( $10^8$ ;  $10^9$ ; the disjoint decade), 3.0–6.9 million pairs per gap at  $10^9$ . No gap crosses the 30-test Bonferroni threshold in any window (max  $|\text{corr}| = 0.00101$ ; per-gap s.e. 0.0003–0.0005); the  $z$ -ensembles are unit-variance noise (std 1.13/0.95/0.96); the class-level  $\chi^2$  totals sit on their dof (475/480, 480/480, 462/480). Meanwhile the same pair counts reproduce the singular-series sawtooth  $\prod_{p|g, p>2} \frac{p-1}{p-2}$  (range 1.00–2.67) point by point: mean measured/HL ratio 0.9989, worst deviation 0.0015 (Fig. 2).

**Affine pairs.** Sophie-Germain pairs  $(p, 2p+1)$  ( $N = 1,775,672$ ,  $p \leq 5 \times 10^8$ ):  $\langle a_p a_q \rangle = -0.0002 \pm 0.0008$  ( $z = -0.28$ ); count against the corresponding  $2C_2 \int dx/(\ln x \ln 2x)$  prediction: ratio 0.9999. Likewise  $(p, 2p-1)$ :  $z = -0.46$ .

**Verdict: uniform blindness at the  $10^{-3}$  level per statistic**, measured against a loudly structured classical background — to our knowledge the first such measurement for a  $\mathrm{PSL}(2, 7)$  extension. One corollary deserves emphasis, since it answers a question we are often asked: the twin-prime constant is an abelian phenomenon, and this maximally non-abelian channel measurably carries none of it. The conductor-21 program is structurally orthogonal to the twin prime conjecture; no route is claimed or implied.

## 6 Chebotarev races predicted by our own zeros

Let  $E'_C(x) = \frac{\ln x}{\sqrt{x}} \left[ \frac{|G|}{|C|} \pi_C(x) - \pi_{\mathrm{unram}}(x) \right]$  for the observable classes  $C \in \{1A, 2A, 4A, 3A, 7\}$  (the septic's cycle types merge  $7A \cup 7B$ , which conveniently makes all class-averaged character weights real). Following Rubinstein–Sarnak [8] and Ng's non-abelian treatment [9]:

- **Bias means (exact character theory; no central zeros, as the tables support):**  $\mu_C = 1 - r_2(C)$  with  $r_2 = 1 + \chi_6 + \chi_7 + \chi_8$ , giving  $(-21, -1, +1, 0, 0)$ . The fully split primes trail enormously (the identity has 22 square roots in  $G$ );  $4A$  — the unique class nothing squares into,  $G$  having no elements of order 8 — is the only class racing ahead.
- **Measured to  $10^9$**  (log-averaged over  $[10^4, 10^9]$ ):  $(-32.9, -1.8, +1.5, -0.0, +0.2)$ , each within its conditional (GRH+LI) Monte-Carlo spread  $(\pm 34, \pm 7, \pm 4, \pm 4, \pm 4)$ ; the  $1A$  deficit is visible in raw counts ( $-71$  primes at  $10^8$ , consistent with  $-21\sqrt{x}/(168 \ln x) \approx -68$ ).
- **Shape from the zero tables:** reconstructing the oscillation from the four low-zero tables alone reproduces the measured races at correlations  $0.81/0.92/0.95/0.94/0.97$  over six decades (Fig. 3) — the first dynamical validation of the companion tables, and a direct exhibition of Rubinstein–Sarnak low-zero dominance (10–15 zeros per character suffice).
- **Bias densities (conditional, first computed for a  $\mathrm{PSL}(2, 7)$  field):**  $P(E'_{1A} < 0) = 0.727$ ; pairwise lead probabilities  $\delta(1A > X) \approx 0.27\text{--}0.29$ ;  $\delta(4A > 3A) = 0.56$ ;  $3A$  vs  $7$  dead even. Note the structural contrast with the classical mod-4 race ( $\delta \approx 0.996$ ): the identity class couples to every character with weight  $\deg \chi$ , so its oscillation variance carries the full column norm  $|G| - 1 = 167$  — *in a strongly non-abelian race the split-prime bias is large in mean but weak in probability.*

## 7 The bulk form factor, measured from primes

The variance of  $\psi_\chi$  in short intervals, computed from primes, probes the pair correlation of the zeros (Goldston–Montgomery [5]; Montgomery–Soundararajan [6]; for the correlation–prime dictionary see also Bogomolny–Keating [7]). Under the stationary cluster expansion (*conditional*: zeros on the critical line plus the cluster/stationarity hypotheses — the measurements below are unconditional),

$$\mathrm{Var} \left[ \frac{\psi_\chi(xe^\delta) - \psi_\chi(x)}{\sqrt{x}} \right] \approx \int_0^\infty \bar{\rho}(\gamma) 2|\varphi_\gamma(\delta)|^2 F(\alpha(\gamma, x)) d\gamma, \quad \alpha = \frac{\ln x}{2\pi\bar{\rho}(\gamma)},$$

with  $|\varphi_\gamma|^2 = (e^\delta - 2e^{\delta/2} \cos \gamma\delta + 1)/(\frac{1}{4} + \gamma^2)$  and  $F$  the Montgomery form factor [4]:  $F = \min(\alpha, 1)$  for GUE,  $F \equiv 1$  for uncorrelated zeros. The decisive regime is  $\alpha < 1$ . For  $L(s, \chi_8)$ ,  $2\pi\bar{\rho} = 15.7 + 8 \ln \gamma$ , so **at  $x = 10^8$  every height  $\gamma \gtrsim 2$  is correlation-sensitive**, and GUE vs. uncorrelated differ by a factor 2–5; for  $\zeta$  at the same  $(x, \delta)$ ,  $\alpha \approx 3\text{--}4$  and the prediction is

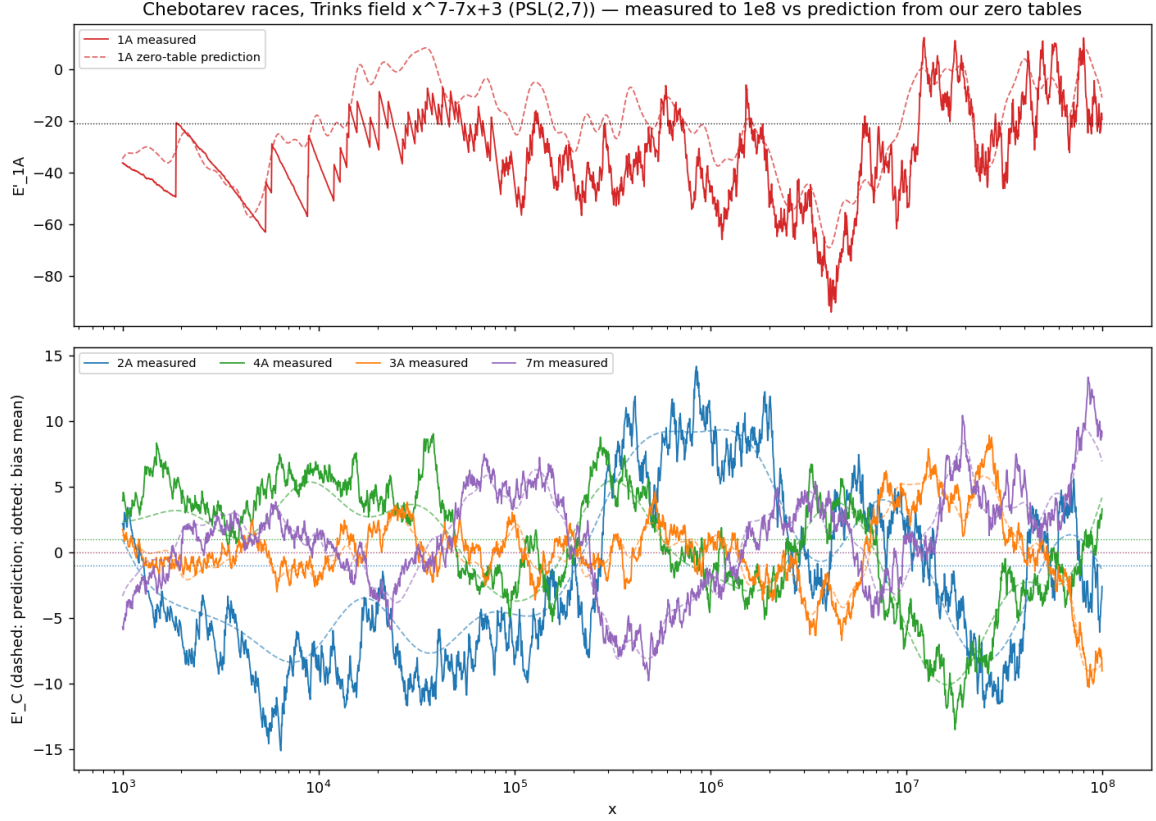


Figure 3: Chebotarev races for the Trinks field to  $10^8$  (extended to  $10^9$  in the data release): measured (solid) vs. reconstruction from our zero tables (dashed); dotted lines at the exact bias means  $1 - r_2(C)$ .

diagonal — the built-in control. In the deep regime the GUE ramp makes the density cancel exactly, giving the parameter-free identity  $\text{Var}_{\text{GUE}} = \delta \ln x$  (primes behaving as uncorrelated signs).

Measured over 26 interval lengths  $\delta \in [10^{-4}, 0.3]$  and windows  $[10^7, 10^8]$ ,  $[10^8, 10^9]$  (exact primes),  $[10^9, 10^{11}]$  (the program’s earlier deep-sweep signal):

| window                                       | $\chi^2$ vs GUE | $\chi^2$ vs uncorrelated          | mean $R = \text{Var}/\text{Var}_{\text{unc}}$ |
|----------------------------------------------|-----------------|-----------------------------------|-----------------------------------------------|
| $[10^7, 10^8]$ , $\delta \leq 0.05$ (20 pts) | 5.5             | 585,788                           | 0.327 (GUE: 0.316)                            |
| $[10^8, 10^9]$ , $\delta \leq 0.05$ (20 pts) | 11.7            | 470,710                           | meas/GUE = 1.009 at $\delta \leq 0.01$        |
| $\zeta$ control (both windows)               |                 | measured/predicted = 0.996, 0.994 |                                               |

The three windows ride three separate GUE curves — the  $\ln x$ -dependence is itself a confirmed prediction (Fig. 4) — and the deep-regime identity lands at bench/GUE = 1.001. Effective coverage:  $\alpha \approx 0.22$ – $0.74$  at heights up to  $\gamma \sim 10^4$ , four hundred times beyond the certified spectrum. **This is the first two-level measurement of this spectrum in the bulk, and it is GUE at the percent level** — consistency with the expected universality class [10], not discovery, and not evidence concerning RH. (At  $\delta \gtrsim 0.15$  the statistic is dominated by the few lowest actual zeros — one realization, under half an oscillation per window — and both the ensemble prediction and the error estimate degrade; that regime is excluded by design.)

Two extensions complete the picture. *Gaussianity*: the skewness and excess kurtosis of the increments match the *exact* shot-noise (diagonal) predictions  $\sum \Lambda_\chi^3/x^{3/2}$ ,  $\sum \Lambda_\chi^4/x^2$  ( $\langle a^3 \rangle =$

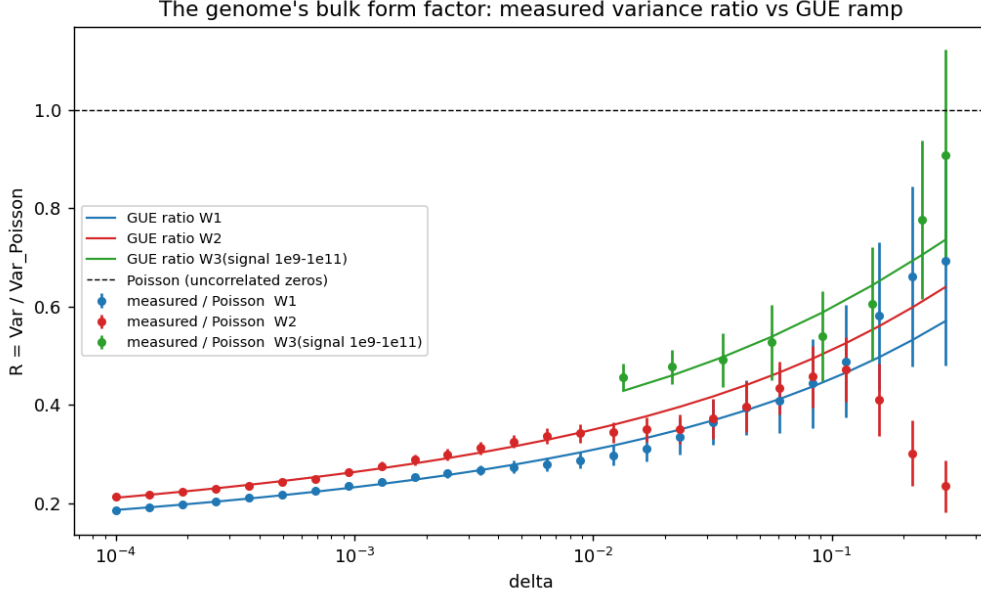


Figure 4: The bulk form factor from primes: measured variance ratio  $R = \text{Var}/\text{Var}_{\text{uncorrelated}}$  vs. the GUE ramp prediction, three windows spanning  $x = 10^7$  to  $10^{11}$ . Uncorrelated zeros ( $R = 1$ ) are excluded at  $\chi^2 \approx 5 \times 10^5$ .

3): a naive  $3.7\sigma$  skewness excess collapses to  $0.85\sigma$  once the shot term is included — third- and fourth-order zero correlations bounded at the Gaussian level. *Scale purity*: the ratio  $\text{Var}/\text{Var}_{\text{GUE}}$  at fixed  $\delta$  is flat across five exact decades  $10^6 \rightarrow 10^{10}$  (weighted slopes  $+0.022 \pm 0.015$  and  $+0.005 \pm 0.018$  per unit  $\ln x$ ); an off-critical-line contribution of exponent  $\beta$  would grow like  $x^{2\beta-1}$ , so this is a prime-only consistency check in the spirit of the program’s earlier prime-side sweeps — corroboration, not proof.

## 8 Flags raised and retired (the error record)

Three fluctuations appeared during the campaign; all three are part of the result. (i) A  $+2.76\sigma$  correlation at gap 4 (one of  $\sim 30$  tests) *flipped sign* to  $-2.26\sigma$  on the pre-registered disjoint decade  $(10^8, 10^9]$ : retired. (ii) A  $3.7\sigma$  skewness excess was the exact, predictable prime shot noise: retired by prediction. (iii) A  $+2.3\sigma$  drift in the purity ratio traced entirely to the coarsest corner of the legacy binned signal; the exact sweep to  $10^{10}$  (404M primes) shows a flat ratio: retired by better data. We consider this discipline — pre-registered retests, exact null predictions before interpretation — as much a part of the companion’s content as the confirmations.

## 9 What this does and does not show

**Does:** the object of [1] behaves, measurably and at scale, exactly as a healthy primitive  $L$ -function should: explicit formula exact; low zeros GUE-consistent with symmetry type  $\text{SO}(\text{even})$ ; bulk pair correlation on the Montgomery ramp at the percent level; Chebotarev races carrying precisely the biases character theory dictates; and a Frobenius channel provably-in-data blind to all pair structure of the primes, as the simplicity of  $\text{PSL}(2, 7)$  suggests it must be. The high conductor, a computational burden elsewhere in the program, here acts as a microscope on the form factor.

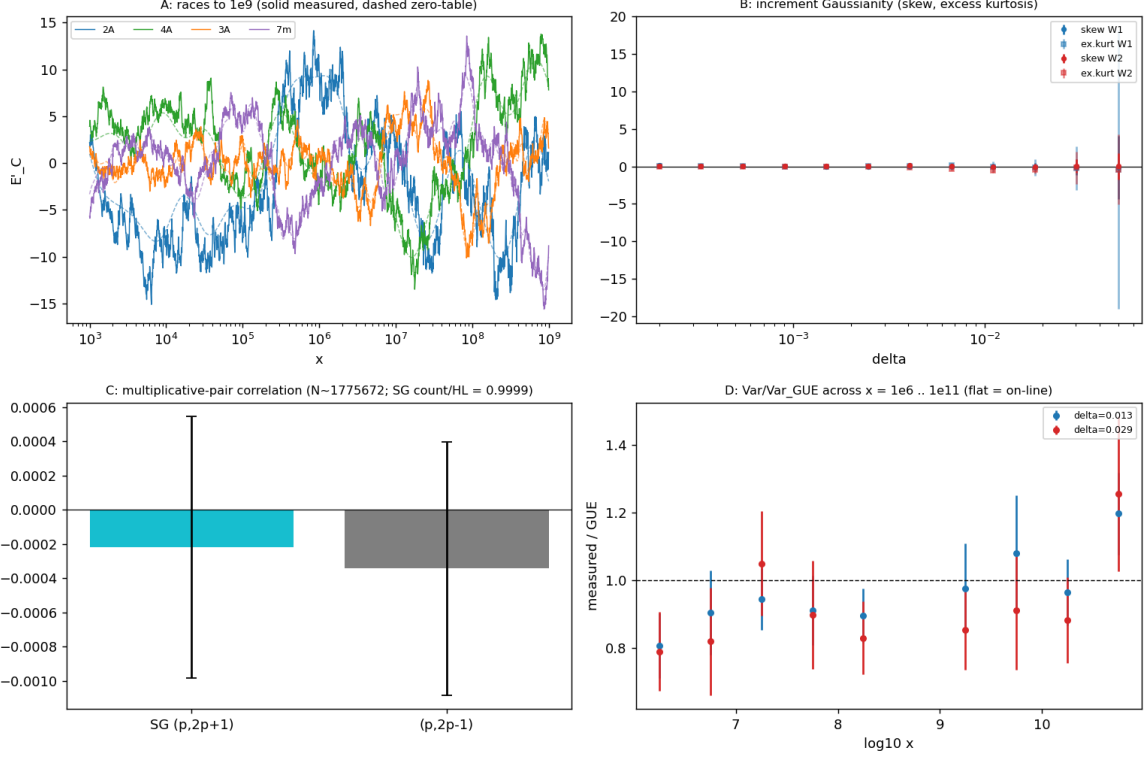


Figure 5: Closing probes: races at  $10^9$ ; increment Gaussianity after exact shot noise; affine-pair blindness with its Hardy–Littlewood count control; purity scaling across  $x$ .

**Does not:** nothing above bears on the Riemann Hypothesis for  $L(s, \chi_8)$  or on the operator-theoretic open problem of the main paper; GUE consistency is the expected behaviour of the universality class, and the conditional statements (race densities, cluster-expansion predictions) are labelled. Nothing here advances the twin prime conjecture; §5 shows the opposite — a measured decoupling.

## Data and code availability

All scripts, zero tables, Frobenius caches (regenerable;  $\leq 10^9$  classes and  $(10^9, 10^{10}]$  binned  $\psi_\chi$ ), figures, and per-run findings notes with SHA-256 seals are released with this deposit; the certified zero data and the main program record are at [2]. Every number in this paper is produced by a named script on named data. The second author’s independent verification covers the theory on which these statistics rest — the bias-mean formula and its square-root counts, the Frobenius–Schur indicators, and the framework identities, re-derived from first principles — and not a rerun of the measurement campaign itself; the scripts and data are released precisely so that any reader can perform that rerun. This paper and its accompanying scripts, findings notes, and data tables are released under the Creative Commons Attribution 4.0 International licence (CC BY 4.0).

## A Model attribution

Following the convention of [1] (Appendix F): the measurement campaigns of §§3,5–7 were designed and executed with Claude (Fable 5, Anthropic) on 2026-09-11; the low-zero statistics of §4 with Claude (Opus) on 2026-08-29; the certified zero tables and engines are those of the

main program. All mathematical claims were verified by computation; responsibility for the content rests with the human authors.

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